



## **CASE STUDY**

# **AI IMPACT ON JOBS: MACHINE LEARNING-BASED LAYOFF RISK PREDICTION**

**BY**

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# CERTIFICATE

This is to certify that the case study entitled “**AI Impact on Jobs: Machine Learning-Based Layoff Risk Prediction**” is a bonafide work carried out by **Sahith Reddy**, bearing Roll No. **160124748128**, of the Department of Artificial Intelligence and Machine Learning, Chaitanya Bharathi Institute of Technology (A), Hyderabad, as part of the Introduction to Machine Learning Lab during the academic year 2025–2026.

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# DECLARATION

I hereby declare that the case study entitled “**AI Impact on Jobs: Machine Learning-Based Layoff Risk Prediction**” is my original work carried out as part of the Introduction to Machine Learning Lab.

The study presents the data exploration, preprocessing, feature engineering, machine learning model development, validation, optimization, and evaluation performed as part of the project *Machine Learning Blueprint for AI Workforce Impact Analysis*.

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# ABSTRACT

Artificial Intelligence (AI) and automation are transforming the global workforce by changing the nature of tasks performed across different industries. While AI can improve productivity and create new opportunities, increasing automation can also expose certain roles to workforce displacement.

This case study presents a machine learning blueprint for analyzing and predicting AI-related workforce impact, with particular emphasis on layoff-risk prediction. The project uses a consolidated dataset containing approximately 23,000 workforce records and develops a structured machine learning workflow covering exploratory data analysis, data cleansing, feature transformation, feature engineering, model development, validation, hyperparameter optimization, and evaluation.

The exploratory analysis revealed that the numerical workforce features showed very weak linear relationships, with the observed correlation structure approaching  $r \approx 0.00$ . This motivated the selection of non-linear ensemble methods, specifically Random Forest and Extreme Gradient Boosting (XGBoost).

Several engineered features were introduced, including `Risk_vs_Adoption_Index`, `Years_Experience` and `Job_Title_Frequency`. The dataset was divided into an 80% development set and a 20% untouched test holdout. Five-fold stratified cross-validation was used during model development and tuning.

Because workforce disruption data can be class-imbalanced, F1-Score was selected as the primary optimization metric, with ROC-AUC used as an important supporting measure.

The documented experimental results show that the optimized XGBoost classifier achieved the strongest performance, with an accuracy of **91.2%**, precision of **90.4%**, recall of **89.1%**, F1-Score of **89.7%**, and ROC-AUC of **95.8%**. The optimized Random Forest achieved an F1-Score of **85.5%** and ROC-AUC of **92.3%**.

The study demonstrates how carefully designed machine learning pipelines can model complex workforce relationships while reducing data leakage, overfitting, and class-imbalance risks.

**Keywords:** Artificial Intelligence, Machine Learning, Workforce Impact, Layoff Risk, Random Forest, XGBoost, Feature Engineering, F1-Score, ROC-AUC.

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# 1. INTRODUCTION

## 1.1 Background

Artificial Intelligence has become an important technological force affecting organizations, industries, and the global workforce.

Modern AI systems can automate repetitive operations, assist decision making, analyze large volumes of information, and improve operational efficiency. As organizations adopt these technologies, some existing tasks may become partially or completely automated.

However, the relationship between AI and employment is not simply a binary problem of whether AI will replace humans. The degree of impact depends on characteristics of individual jobs and industries.

Factors such as routine task percentage, automation level, creativity requirements, human interaction, job level, industry, and economic conditions can influence the exposure of a role to automation.

Therefore, machine learning can be used to identify complex relationships among these variables and estimate the associated layoff-risk category.

## 1.2 Project Overview

The project is titled:

### **Machine Learning Blueprint for AI Workforce Impact Analysis**

The objective is to develop a structured machine learning workflow for analyzing AI's multi-dimensional impact on the workforce.

The complete workflow progresses through:

Planning → EDA → Preprocessing → Feature Engineering → Model Development → Validation → Op  
(1.1)

### 1.3 Motivation

The project is motivated by the need for a data-driven approach to understanding AI-related workforce disruption.

A predictive system can potentially assist with:

- Workforce risk assessment.
- Reskilling and upskilling planning.
- Identification of highly automatable roles.
- Organizational workforce planning.
- Career planning.
- Human-AI collaboration analysis.

The purpose of the model is not to determine that an individual will definitely lose employment. Instead, it provides a predictive estimate based on patterns learned from the available data.

### 1.4 Problem Statement

The problem addressed by this case study is:

*Given a set of workforce and job-related characteristics, predict the associated layoff-risk indicator using supervised machine learning.*

The target variable is:

`Layoff_Risk_Indicator`

The remaining relevant variables are treated as predictive features.

### 1.5 Research Question

The primary research question is:

*Which machine learning approach can most effectively model the complex and non-linear relationships between workforce characteristics, AI adoption, automation, and layoff risk?*

## 2. OBJECTIVES

The major objectives of the project are:

1. To explore workforce data and identify important statistical patterns.
2. To identify missing values, duplicate records, anomalies, and extreme observations.
3. To develop a robust data-cleaning and preprocessing strategy.
4. To transform categorical and continuous variables into machine-learning-compatible representations.
5. To engineer features that capture interactions between automation and workforce characteristics.
6. To develop Random Forest and XGBoost classification models.
7. To establish a simple Decision Tree baseline.
8. To implement stratified cross-validation while preventing information leakage.
9. To optimize model hyperparameters using Randomized Search followed by Grid Search.
10. To evaluate models using Precision, Recall, F1-Score, and ROC-AUC.
11. To analyze overfitting and underfitting risks.
12. To identify important workforce characteristics contributing to predictions.

## 3. DATASET DESCRIPTION

### 3.1 Dataset Overview

The project works with a consolidated workforce dataset containing approximately:

**23,000 workforce records**

The dataset is designed to represent multiple dimensions of the impact of AI and automation on employment.

The project reports reference public workforce datasets including AI Impact on Jobs 2030 and AI Impact Jobs Layoff Risk repositories.

### 3.2 Target Variable

The target variable used in the final modeling strategy is:

`Layoff_Risk_Indicator`

This variable represents the layoff-risk outcome that the classification models attempt to predict.

### 3.3 Feature Categories

The dataset contains a mixture of continuous, categorical, ordinal, and high-cardinality variables.

Important variables discussed throughout the project include:

Table 3.1: Important Workforce Variables

| Feature                    | Role / Interpretation                                        |
|----------------------------|--------------------------------------------------------------|
| Years_Experience           | Employee or worker experience level                          |
| Average_Salary_USD         | Average compensation associated with a role                  |
| Work_Hours_Per_Week        | Typical weekly working hours                                 |
| Routine_Task_Percentage    | Proportion of work consisting of routine tasks               |
| Tasks_Automated_Percentage | Degree to which tasks can be automated                       |
| AI_Adoption_Level          | Level of AI adoption associated with the role or environment |
| AI_Replacement_Risk        | Potential exposure to AI-driven replacement                  |
| Future_Demand_Score        | Expected future demand associated with a role                |
| Creativity_Requirement     | Degree of creativity required by the job                     |
| Human_Interaction_Level    | Level of interpersonal interaction required                  |
| Job_Level                  | Seniority or responsibility level                            |
| Industry                   | Industrial sector associated with the role                   |
| Job_Title                  | Occupation or job title                                      |

### 3.4 Non-Predictive Attributes

Attributes such as **Employee\_ID** are considered identifiers rather than meaningful predictive variables.

Such columns should be removed before model training to prevent the model from learning irrelevant patterns.

## 4. EXPLORATORY DATA ANALYSIS

### 4.1 Purpose of EDA

Exploratory Data Analysis (EDA) was performed to understand the structure, distribution, relationships, and anomalies within the workforce data.

The major objectives were:

- Understand numerical feature distributions.
- Identify categorical structures.
- Detect missing values.
- Detect duplicate records.
- Identify extreme observations.
- Study relationships between variables.
- Determine whether linear models would be suitable.

### 4.2 Univariate Analysis

Initial inspection was performed using operations such as:

```
df.head()  
df.info()  
df.describe()
```

These operations provide information about:

- Dataset dimensions.
- Data types.
- Summary statistics.
- Feature distributions.
- Potential missing values.

### 4.3 Bivariate Analysis

Several relationships were identified as useful for investigation.

Examples include:

- Years of Experience versus Average Salary.
- AI Replacement Risk versus Future Demand Score.
- Performance Score versus Job Satisfaction.
- AI Adoption Level versus Layoff Risk Indicator.

These comparisons help identify trends, clusters, nonlinear patterns, and potential anomalies.

### 4.4 Visualization Techniques

The project uses four major visualization techniques.

Table 4.1: EDA Visualization Strategy

| Visualization        | Application                                                    | Expected Outcome                                       |
|----------------------|----------------------------------------------------------------|--------------------------------------------------------|
| Histograms with KDE  | Continuous features such as salary and routine-task percentage | Identify distributions, skewness, and concentration    |
| Box Plots            | Continuous variables across categories                         | Identify outliers and compare class variance           |
| Scatter Plots        | Pairwise continuous variables                                  | Identify trends, clusters, and nonlinear relationships |
| Correlation Heatmaps | Numerical feature matrix                                       | Identify correlation and potential collinearity        |

### 4.5 Correlation Analysis

Correlation analysis was used to study linear relationships between numerical variables.

A key observation reported during EDA was that the raw continuous features exhibited approximately zero linear correlation:

$$r \approx 0.00 \quad (4.1)$$

This suggested that the underlying relationships were highly orthogonal or nonlinear.

This finding became an important reason for selecting tree-based ensemble algorithms rather than relying exclusively on traditional linear models.

## 5. DATA CLEANING AND PREPROCESSING

### 5.1 Missing Value Analysis

Missing values were categorized according to their potential mechanism.

#### 5.1.1 Missing Completely at Random

For Missing Completely at Random (MCAR), row-wise deletion can be considered when the proportion of missing observations is sufficiently small.

The project specifies a threshold of less than 2% of the 23,000-record dataset.

#### 5.1.2 Missing at Random

For Missing at Random (MAR), missingness may depend on another observed feature.

For example, missing salary information could depend on industry.

The proposed strategy is grouped median imputation:

$$x_{\text{missing}} = \text{Median}(x \mid \text{group}) \quad (5.1)$$

This preserves subgroup characteristics better than global mean imputation.

#### 5.1.3 Missing Not at Random

For Missing Not at Random (MNAR), a missingness indicator can be introduced before imputation.

For example:

|                                |
|--------------------------------|
| <code>Feature_IsMissing</code> |
|--------------------------------|

This allows tree-based models to potentially learn information contained in the missingness pattern.

### 5.2 Duplicate Records

Duplicate observations can introduce bias and distort statistical analysis.

Duplicate detection can be performed using:

```
df.duplicated()
```

If duplicates are present, they should be removed so that each observation is represented only once.

### 5.3 Outlier Detection

Potential outlier variables include:

- Average\_Salary\_USD
- Years\_Experience
- Work\_Hours\_Per\_Week
- Job\_Growth\_2030

The project considers:

- Box plots.
- Z-score analysis.
- Interquartile Range (IQR).

For the IQR method:

$$IQR = Q_3 - Q_1 \quad (5.2)$$

The lower and upper limits are:

$$Lower = Q_1 - 1.5(IQR) \quad (5.3)$$

$$Upper = Q_3 + 1.5(IQR) \quad (5.4)$$

### 5.4 Winsorization

Rather than removing legitimate extreme observations, the capstone workflow applies Winsorization at the 99th percentile for extreme continuous observations.

This is particularly relevant for high-salary technology cohorts, where extreme values may represent valid observations rather than data-entry errors.

## 5.5 Feature Scaling

The project considers two scaling strategies.

### 5.5.1 Standardization

Standardization transforms a variable using:

$$z = \frac{x - \mu}{\sigma} \quad (5.5)$$

It is applied to continuous variables such as:

- Years\_Experience
- Average\_Salary\_USD
- Work\_Hours\_Per\_Week

### 5.5.2 Min-Max Normalization

For bounded variables:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (5.6)$$

This maps values to a bounded interval, typically:

$$[0, 1]$$

## 6. CATEGORICAL VARIABLE ENCODING

### 6.1 Ordinal Encoding

Ordinal variables contain an inherent order.

Examples include:

- Education\_Level
- Automation\_Level

These can be represented using sequential integer mappings.

### 6.2 One-Hot Encoding

Low-cardinality nominal variables can be transformed using One-Hot Encoding.

Examples include:

- Employment Type
- Industry
- Country

For linear models, the first category can be removed to avoid the dummy-variable trap.

### 6.3 High-Cardinality Encoding

Variables such as `Job_Title` can have many unique categories.

Direct One-Hot Encoding may therefore generate an unnecessarily large feature matrix.

The project considers Target/Mean Encoding for high-cardinality categorical variables.

The critical engineering requirement is that target encoding must be performed within cross-validation folds.

If target statistics are calculated globally before cross-validation, validation information can leak into the training features and produce artificially high validation scores.

Therefore:

$$\text{Target Encoding} \rightarrow \text{Internal CV Only} \tag{6.1}$$

## 7. FEATURE ENGINEERING

Feature engineering was used to expose relationships that may not be directly represented by the original variables.

### 7.1 Risk versus Adoption Index

The first engineered feature is:

$$Risk\_vs\_Adoption\_Index = Routine\_Task\_Percentage \times AI\_Adoption\_Level \quad (7.1)$$

This interaction captures roles where routine work is high while AI adoption is simultaneously increasing.

Such roles may represent higher exposure to automation.

### 7.2 Years Experience Squared

The second engineered feature is:

$$Years\_Experience\_Sq = Years\_Experience^2 \quad (7.2)$$

The purpose is to capture possible nonlinear effects of experience.

For example, the influence of additional experience may not remain constant across all career stages.

### 7.3 Job Title Frequency

The third engineered feature is:

$$Job\_Title\_Frequency$$

This represents the relative occurrence of a job title in the dataset.

It provides information about occupational density before model training.

## 7.4 Feature Selection

The capstone workflow proposes combining:

- Variance Inflation Factor (VIF).
- Embedded feature selection.
- `SelectFromModel`.
- Regularized XGBoost.

VIF can be used to identify redundant or highly collinear variables introduced during feature engineering.

## 8. MODEL DEVELOPMENT

### 8.1 Model Selection Rationale

The EDA indicated that the workforce dataset contains complex nonlinear interactions.

Consequently, the project focuses on two ensemble architectures:

1. Random Forest Classifier.
2. Extreme Gradient Boosting (XGBoost).

A Decision Tree stump is used as a simple baseline.

### 8.2 Decision Tree Baseline

The baseline model is a simplified Decision Tree with:

`max_depth = 3`

The purpose of the baseline is to establish a minimum performance reference against which more advanced models can be compared.

### 8.3 Random Forest

Random Forest is an ensemble learning method based on Bootstrap Aggregating (Bagging).

It trains multiple decision trees using:

- Bootstrap samples.
- Random feature subsets.
- Independent tree construction.

The final classification is determined through majority voting:

$$\hat{y} = \text{mode} \{T_1(x), T_2(x), \dots, T_B(x)\} \quad (8.1)$$

Random Forest is suitable for the workforce problem because it can capture nonlinear interactions between variables such as salary, automation, and routine task percentage.

### 8.3.1 Advantages

- Robust to nonlinear relationships.
- Scale-invariant for tree splits.
- Resistant to individual-tree overfitting.
- Provides feature importance.

### 8.3.2 Potential Limitations

- High memory consumption for large ensembles.
- Computational cost increases with tree complexity.
- Deep trees may increase variance.

## 8.4 XGBoost

Extreme Gradient Boosting constructs trees sequentially.

Unlike Random Forest, where trees are built independently, each new XGBoost tree attempts to correct errors made by the previous ensemble.

A simplified representation is:

$$F_m(x) = F_{m-1}(x) + \eta h_m(x) \quad (8.2)$$

where:

- $F_m(x)$  is the updated ensemble.
- $\eta$  is the learning rate.
- $h_m(x)$  is the newly trained tree.

XGBoost also incorporates regularization through  $L_1$  and  $L_2$  penalties.

### 8.4.1 Advantages

- Strong performance on structured tabular data.
- Captures nonlinear feature interactions.
- Built-in regularization.
- Efficient implementation.

### 8.4.2 Potential Limitations

- Sensitive to hyperparameter selection.
- Can overfit without regularization.
- Requires careful tuning of tree depth and learning rate.

## 9. DATA PARTITIONING AND VALIDATION

### 9.1 Train-Test Partition

The complete dataset of approximately 23,000 records is divided into:

$$80\% = \text{Development Set} \tag{9.1}$$

$$20\% = \text{Test Holdout} \tag{9.2}$$

The split is stratified according to:

`Layoff_Risk_Indicator`

This preserves the target-class distribution.

### 9.2 Test Set Isolation

The 20% test set is isolated immediately after the initial split.

It must not be used for:

- Hyperparameter optimization.
- Fitting preprocessing transformations.
- Threshold calibration.
- Feature-selection decisions.

It serves as an independent final evaluation set.

### 9.3 Five-Fold Stratified Cross-Validation

The remaining 80% development set is evaluated using five-fold stratified cross-validation.

In each iteration:

- Four folds are used for training.

- One fold is used for validation.

The process is repeated five times.

The average validation score can be represented as:

$$\mu = \frac{1}{5} \sum_{i=1}^5 Score_i \quad (9.3)$$

The standard deviation is also monitored to identify unstable models.

The Week 4 framework specifies that:

$$\sigma > 0.03 \quad (9.4)$$

can be treated as an instability signal requiring further pipeline investigation.

## 10. EVALUATION METRICS

### 10.1 Why Accuracy Alone Is Insufficient

Workforce disruption data may contain class imbalance because stable employment states can substantially outnumber immediate layoff-risk events.

In such a situation, a model could obtain high accuracy by primarily predicting the majority class.

Therefore, the project prioritizes F1-Score and ROC-AUC.

### 10.2 Precision

Precision measures the proportion of predicted positive observations that are actually positive:

$$Precision = \frac{TP}{TP + FP} \quad (10.1)$$

High precision reduces false alarms.

### 10.3 Recall

Recall measures the proportion of actual positive cases successfully identified:

$$Recall = \frac{TP}{TP + FN} \quad (10.2)$$

High recall reduces missed high-risk cases.

### 10.4 F1-Score

F1-Score is the harmonic mean of precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (10.3)$$

F1-Score is the primary optimization metric in the project because it balances false positives and false negatives.

## 10.5 ROC-AUC

Receiver Operating Characteristic Area Under the Curve (ROC-AUC) measures the model's ability to discriminate between classes across different probability thresholds.

A higher ROC-AUC indicates stronger discriminatory capability.

## 10.6 Accuracy

Accuracy is still reported for completeness:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (10.4)$$

However, it is not treated as the primary optimization criterion.

## 11. CLASS IMBALANCE AND THRESHOLD CALIBRATION

### 11.1 Class Imbalance

Workforce datasets may contain considerably more stable employment records than high-risk layoff observations.

A model trained without considering this imbalance may develop a majority-class bias.

### 11.2 XGBoost Class Weighting

For XGBoost, the project considers:

```
scale_pos_weight
```

This parameter adjusts the relative contribution of positive-class observations.

### 11.3 Random Forest Class Weighting

For Random Forest, the project specifies:

```
class_weight='balanced_subsample'
```

This adjusts class weighting within bootstrap samples.

### 11.4 Probability Threshold Calibration

A default classification threshold of 0.5 may not be optimal for imbalanced classification.

Therefore, the project proposes generating a Precision-Recall curve and identifying a threshold that maximizes F1-Score.

The process is:

$$P(y = 1|x) \rightarrow \text{Precision-Recall Curve} \rightarrow \text{Optimal Threshold} \rightarrow \text{Maximum F1} \quad (11.1)$$

This allows the decision threshold to reflect the practical requirements of workforce-risk classification.

## 12. HYPERPARAMETER OPTIMIZATION

### 12.1 Two-Stage Optimization

The project follows a two-stage optimization strategy.

#### 12.1.1 Stage 1: Randomized Search

RandomizedSearchCV explores a broad hyperparameter space using approximately 50 sampled configurations.

This efficiently identifies promising regions without evaluating every possible combination.

#### 12.1.2 Stage 2: Grid Search

After identifying a promising region, GridSearchCV performs an exhaustive search over a smaller localized parameter grid.

Therefore:

$$\text{Broad Random Search} \rightarrow \text{Promising Region} \rightarrow \text{Fine Grid Search} \quad (12.1)$$

### 12.2 Random Forest Search Space

The documented Random Forest search space includes:

Table 12.1: Random Forest Hyperparameters

| Parameter         | Values        |
|-------------------|---------------|
| n_estimators      | 100, 300, 500 |
| max_depth         | 10, 20, None  |
| min_samples_split | 2, 5, 10      |

### 12.3 XGBoost Search Space

The documented XGBoost search space includes:

Table 12.2: XGBoost Hyperparameters

| Parameter     | Values          |
|---------------|-----------------|
| learning_rate | 0.01, 0.05, 0.1 |
| max_depth     | 3, 5, 7         |
| subsample     | 0.7, 0.8, 1.0   |
| reg_lambda    | 1, 10, 50       |

The project also documents an expanded XGBoost search space during model-development experimentation:

- Learning rate: 0.01, 0.05, 0.1, 0.2
- Maximum depth: 3, 6, 9
- $L_2$  regularization: 1, 5, 10

## 12.4 Early Stopping

The evaluation framework monitors validation performance and can stop XGBoost training when validation loss fails to improve.

The documented configuration uses a patience of approximately 10 consecutive iterations.

This helps reduce unnecessary training and limits overfitting.

## 13. MODEL PERFORMANCE

### 13.1 Experimental Results

The documented Week 3 model-development results are summarized below.

Table 13.1: Model Performance Comparison

| Model                   | Accuracy     | Precision    | Recall       | F1           | ROC-AUC      |
|-------------------------|--------------|--------------|--------------|--------------|--------------|
| Decision Tree Stump     | 0.642        | 0.589        | 0.531        | 0.558        | 0.614        |
| Optimized Random Forest | 0.874        | 0.861        | 0.849        | 0.855        | 0.923        |
| Optimized XGBoost       | <b>0.912</b> | <b>0.904</b> | <b>0.891</b> | <b>0.897</b> | <b>0.958</b> |

### 13.2 Baseline Performance

The Decision Tree stump achieved:

$$F1 = 55.8\% \tag{13.1}$$

and:

$$ROC-AUC = 61.4\% \tag{13.2}$$

This establishes a relatively weak baseline.

### 13.3 Random Forest Performance

The optimized Random Forest achieved:

$$Accuracy = 87.4\% \tag{13.3}$$

$$Precision = 86.1\% \tag{13.4}$$

$$Recall = 84.9\% \tag{13.5}$$

$$F1 = 85.5\% \tag{13.6}$$

$$ROC-AUC = 92.3\% \tag{13.7}$$

This represents a substantial improvement over the Decision Tree baseline.

## 13.4 XGBoost Performance

The optimized XGBoost model achieved the strongest documented performance:

$$Accuracy = 91.2\% \tag{13.8}$$

$$Precision = 90.4\% \tag{13.9}$$

$$Recall = 89.1\% \tag{13.10}$$

$$F1 = 89.7\% \tag{13.11}$$

$$ROC-AUC = 95.8\% \tag{13.12}$$

Therefore, based on the documented evaluation results, XGBoost is the best-performing model.

## 13.5 Performance Interpretation

The improvement from the Decision Tree baseline to the optimized ensemble models demonstrates the value of:

- Nonlinear modeling.
- Ensemble learning.
- Feature engineering.
- Hyperparameter optimization.
- Class-imbalance handling.
- Cross-validation.

The higher ROC-AUC of XGBoost indicates strong class discrimination, while its F1-Score demonstrates a better balance between precision and recall.

## 14. FEATURE IMPORTANCE AND INTERPRETABILITY

### 14.1 Random Forest Feature Importance

Random Forest provides embedded feature-importance estimates.

The project calculates feature importance conceptually using:

```
importance = pd.DataFrame({
    "Feature": X.columns,
    "Importance": best_rf.feature_importances_
})

importance = importance.sort_values(
    by="Importance",
    ascending=False
)
```

The resulting importance scores indicate the relative contribution of features to the Random Forest decision process.

### 14.2 Important Workforce Characteristics

The project identifies the following characteristics as important to understanding layoff-risk predictions:

- **Routine Task Percentage:** A higher proportion of routine work may indicate greater potential for automation.
- **Tasks Automated Percentage:** Indicates the extent to which the tasks associated with a role can already be automated.
- **Creativity Requirement:** Creative requirements can influence how easily a role can be automated.

- **Job Level:** Seniority and responsibility can affect the type and complexity of work performed.
- **Industry:** AI adoption and automation vary across industries.
- **Human Interaction Level:** Jobs involving substantial interpersonal interaction may have different automation exposure.

### 14.3 Feature Importance Visualization

If the actual feature-importance plot has been generated from the Python notebook, it can be inserted here.

**Important:** The numerical importance values should be taken directly from the notebook output. They should not be manually invented.

To insert the actual plot, save it as:

`feature_importance.png`

and uncomment the following block:

## 15. OVERFITTING AND UNDERFITTING ANALYSIS

### 15.1 Overfitting

Overfitting occurs when a model learns training-specific noise rather than generalizable patterns.

Typical indicators include:

- Very high training performance.
- Significantly lower validation performance.
- Declining validation F1-Score.

Potential causes include excessive tree depth, excessive model complexity, and insufficient regularization.

### 15.2 Overfitting Mitigation

The project proposes:

1. Early stopping for XGBoost.
2.  $L_1$  and  $L_2$  regularization.
3. Limiting tree depth.
4. Feature pruning.
5. Strict cross-validation.

### 15.3 Underfitting

Underfitting occurs when a model is too simple to capture the underlying relationships.

Indicators include:

- Poor training performance.
- Poor validation performance.
- Failure to outperform the baseline.

## 15.4 Underfitting Mitigation

Possible approaches include:

1. Increasing model complexity.
2. Increasing tree counts.
3. Adding engineered features.
4. Relaxing excessive regularization.

## 16. TECHNICAL CHALLENGES AND LESSONS LEARNED

### 16.1 High-Cardinality Target Encoding

One major challenge was the possibility of target leakage when encoding high-cardinality categorical variables.

If target encoding is calculated across the complete training dataset before validation, information from validation folds can influence the encoded features.

This can artificially inflate validation performance.

**Lesson Learned:**

Target encoding must be calculated strictly within internal cross-validation folds using out-of-fold target statistics and suitable smoothing.

### 16.2 Hyperparameter Explosion

Large exhaustive parameter grids can create significant computational bottlenecks.

Testing every combination across multiple hyperparameters and folds can become expensive.

**Lesson Learned:**

A two-stage tuning strategy is more efficient:

$$RandomizedSearchCV \rightarrow GridSearchCV \quad (16.1)$$

### 16.3 Probability Threshold Selection

The default probability threshold of 0.5 may not provide the best F1-Score when classes are imbalanced.

**Lesson Learned:**

The classification threshold should be treated as a tunable parameter and optimized using Precision-Recall analysis.

## 16.4 Orthogonal Feature Relationships

The very weak linear correlation observed during EDA demonstrated that simple correlation-based reasoning could miss important relationships.

**Lesson Learned:**

Nonlinear ensemble models are appropriate when predictive information is distributed through interactions rather than strong pairwise linear correlations.

## 17. DEPLOYMENT AND REPRODUCIBILITY

### 17.1 End-to-End Pipeline

A production implementation should bundle all preprocessing and prediction operations into one reproducible pipeline.

The desired architecture is:

Raw Input → Cleaning → Encoding → Scaling → Feature Engineering → Model → Prediction  
(17.1)

This avoids manual transformation inconsistencies during deployment.

### 17.2 Pipeline Serialization

The capstone recommends serializing the complete pipeline using Scikit-learn's Pipeline framework or joblib.

This allows the same transformations applied during training to be automatically reproduced during inference.

### 17.3 Version Control

For reproducibility, the following artifacts should be preserved:

- Raw dataset.
- Preprocessed dataset.
- Encoders.
- Scalers.
- Feature-engineering logic.
- Trained model.
- Hyperparameter configuration.

- Evaluation results.
- Generated visualizations.

## **18. LIMITATIONS**

### **18.1 Dataset Dependency**

The quality and representativeness of the dataset directly affect model performance.

A model trained on a limited workforce dataset may not generalize perfectly to all industries, countries, or employment environments.

### **18.2 Prediction Does Not Imply Causation**

The model identifies statistical patterns associated with layoff risk.

It does not establish that a particular feature directly causes layoffs.

For example, a high automation percentage may be associated with higher risk, but the model cannot by itself prove that automation caused a specific employment decision.

### **18.3 Changing Labor Markets**

AI technology, economic conditions, organizational strategies, and labor markets change continuously.

Therefore, models may require periodic retraining.

### **18.4 Ethical Considerations**

Workforce-risk predictions can influence organizational decisions.

Incorrect or poorly interpreted predictions could lead to:

- Unnecessary restructuring.
- Incorrect employee classification.
- Biased workforce decisions.
- Unjustified employment actions.

Consequently, such models should be used as decision-support tools rather than autonomous decision makers.

## 19. FUTURE SCOPE

### 19.1 SMOTE-NC

The capstone recommends investigating **SMOTE-NC (Synthetic Minority Over-sampling Technique for Nominal and Continuous features)**.

It can be applied within cross-validation to expand the minority decision space while maintaining the distinction between nominal and continuous variables.

### 19.2 SHAP Explainability

Future versions can integrate SHAP (*SHapley Additive exPlanations*) to explain individual model predictions.

A SHAP-based system could answer questions such as:

Which features contributed most strongly to this particular layoff-risk prediction?

This would make the system more transparent than a purely black-box prediction engine.

### 19.3 Interactive Application

The trained model could be deployed through an interactive web application.

A user could provide workforce characteristics and receive:

- Predicted risk category.
- Confidence/probability information.
- Major contributing features.
- Potential automation exposure.

## 19.4 Reskilling Recommendations

A future system could move beyond prediction and recommend:

- Skills to acquire.
- Training programs.
- Alternative career paths.
- Human-AI collaboration opportunities.

## 19.5 Continuous Model Updating

Because AI adoption and employment conditions change over time, the system could be extended to support periodic retraining using newer workforce data.

## 19.6 Human-AI Collaboration Analysis

Instead of measuring only the probability of job replacement, future research could classify jobs into:

1. Low automation exposure.
2. Moderate automation exposure.
3. High automation exposure.
4. Human-AI collaboration opportunities.

## 20. CONCLUSION

This case study presented a complete machine learning blueprint for analyzing AI-related workforce impact and predicting layoff risk.

The project progressed systematically from exploratory data analysis and data quality assessment to preprocessing, feature engineering, model development, validation, optimization, and evaluation.

The analysis of approximately 23,000 workforce records revealed very weak linear correlations among the continuous features, suggesting that important predictive relationships were nonlinear and distributed across feature interactions.

To capture these relationships, Random Forest and XGBoost were selected as the primary ensemble algorithms.

Three important engineered variables were introduced:

- `Risk_vs_Adoption_Index`
- `Years_Experience_Sq`
- `Job_Title_Frequency`

The dataset was divided into an 80% development set and a 20% untouched test set. Five-fold stratified cross-validation was used during development.

Because of the potential class imbalance in workforce-risk prediction, F1-Score was selected as the primary optimization metric, with ROC-AUC used as a major supporting metric.

The documented results demonstrate that the optimized XGBoost model performed best:

$$F1 = 89.7\% \tag{20.1}$$

and:

$$ROC-AUC = 95.8\% \tag{20.2}$$

with an accuracy of:

$$91.2\% \tag{20.3}$$

The optimized Random Forest achieved an F1-Score of 85.5% and ROC-AUC of 92.3%, while the Decision Tree stump baseline achieved an F1-Score of 55.8%.

The project also highlighted important machine learning engineering lessons concerning target-encoding leakage, hyperparameter-search efficiency, class imbalance, and probability-threshold calibration.

Overall, the study demonstrates that a carefully designed machine learning pipeline can provide useful data-driven insights into AI's potential impact on the workforce. However, predictions should be interpreted responsibly because employment outcomes are influenced by economic, organizational, technological, and human factors that cannot be completely captured by a predictive model.

## REFERENCES

1. Week 2 Task Report: *Data Exploration and Preprocessing Strategy*, Machine Learning Blueprint for AI Workforce Impact Analysis, 2026.
2. Week 3 Task Report: *Model Development, Feature Engineering, and Training Strategy*, Machine Learning Blueprint for AI Workforce Impact Analysis, June 25, 2026.
3. Week 4 Task Report: *Model Evaluation and Performance Optimization*, Machine Learning Blueprint for AI Workforce Impact Analysis, 2026.
4. Week 5 Capstone Report: *Comprehensive Project Report and Lessons Learned*, Machine Learning Blueprint for AI Workforce Impact Analysis, July 10, 2026.
5. Scikit-learn Documentation, *Machine Learning in Python*.
6. Scikit-learn Documentation, *Random Forest Classifier*.
7. Scikit-learn Documentation, *RandomizedSearchCV*.
8. XGBoost Documentation, *XGBoost Python API*.

## A. IMPLEMENTATION FRAMEWORK

This appendix summarizes the main implementation structure used in the project. The exact notebook implementation should be attached or included separately if required by the laboratory submission format.

### A.1 Dataset Loading

```
import pandas as pd
import numpy as np

df = pd.read_csv(
    "ai-impact-jobs-layoff-risk-dataset.csv"
)

print(df.head())
print(df.info())
print(df.describe())
```

### A.2 Data Quality Checks

```
# Missing values
print(df.isnull().sum())

# Duplicate records
print(df.duplicated().sum())
```

### A.3 Feature Engineering

```
df["Risk_vs_Adoption_Index"] = (
```

```
df["Routine_Task_Percentage"]  
    * df["AI_Adoption_Level"]  
)  
  
df["Years_Experience_Sq"] = (  
    df["Years_Experience"] ** 2  
)  
  
df["Job_Title_Frequency"] = (  
    df.groupby("Job_Title")["Job_Title"]  
        .transform("count")  
)
```

## A.4 Train-Test Split

```
from sklearn.model_selection import train_test_split  
  
X = df.drop(  
    "Layoff_Risk_Indicator",  
    axis=1  
)  
  
y = df["Layoff_Risk_Indicator"]  
  
X_dev, X_test, y_dev, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.20,  
    stratify=y,  
    random_state=42  
)
```

## A.5 Random Forest

```
from sklearn.ensemble import RandomForestClassifier  
  
rf = RandomForestClassifier(  

```

```
class_weight="balanced_subsample",  
random_state=42  
)
```

## A.6 XGBoost

```
from xgboost import XGBClassifier  
  
xgb = XGBClassifier(  
    random_state=42  
)
```

## A.7 Evaluation

```
from sklearn.metrics import (  
    accuracy_score,  
    precision_score,  
    recall_score,  
    f1_score,  
    roc_auc_score,  
    classification_report,  
    confusion_matrix  
)  
  
y_pred = model.predict(X_test)  
  
accuracy = accuracy_score(  
    y_test,  
    y_pred  
)  
  
precision = precision_score(  
    y_test,  
    y_pred,  
    average="weighted"  
)
```

```
recall = recall_score(
    y_test,
    y_pred,
    average="weighted"
)

f1 = f1_score(
    y_test,
    y_pred,
    average="weighted"
)

print(classification_report(
    y_test,
    y_pred
))
```

## A.8 Feature Importance

```
importance = pd.DataFrame({
    "Feature": X.columns,
    "Importance": best_rf.feature_importances_
})

importance = importance.sort_values(
    by="Importance",
    ascending=False
)

print(importance)
```

## A.9 Feature Importance Plot

The following code can be used to generate the image referenced in Chapter 14:

```
import matplotlib.pyplot as plt

plt.figure(figsize=(10, 6))
```

```
plt.barh(
    importance["Feature"],
    importance["Importance"]
)

plt.xlabel("Importance")
plt.ylabel("Feature")

plt.title(
    "Feature_Importance_-_Optimized_Random_Forest"
)

plt.gca().invert_yaxis()

plt.tight_layout()

plt.savefig(
    "feature_importance.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```